Exact State Alphabets for Blind One-Write Memory

An Erasure Plateau, Sharp Cubical Constructions, and Verified Finite Obstructions

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Abstract

We study the minimum cardinality of a persistent state alphabet used by a deterministic encoder to maintain a binary record under a single blind coordinate assignment, while a phase-aware decoder must recover erased coordinates with zero error. A first exact matching normal form characterizes the high-erasure regime via forced successor edges and residual graph matching. At erasure allowance $q = n - 2$, an exact variational formula over antipodal source sets reduces the problem to an exterior Hamming-neighborhood overlap statistic. A geometric converse, Hamming-prefix cubical packings, and a computer-assisted finite certificate together yield an exact two-level plateau: for $n \geq 7$ and $q \in \{n-3, n-2\}$, the minimum alphabet size is $2^n - \lfloor 2^{n-1}/n \rfloor + \mathbf{1}_{\{n=7\}}$. The exceptional seven-coordinate four-erasure lower bound is supported by five separately checked DRUP refutations of frozen CNF files and a directly audited 120-label protocol. For every $n \geq 8$, matching analytic bounds and explicit constructions establish the formula without finite SAT assumptions. We distinguish label cardinality from rounded bit cost, specify the precise scope of the finite certificate, and give reproducible artifacts for the combinatorial claims.

Keywords: zero-error coding; online state memory; blind overwrite; Hamming graphs; exact combinatorial optimization; computer-assisted proof.

1 Introduction

A blind coordinate assignment discloses its position and new bit to the update operation while concealing the overwritten bit. An encoder that stores only a finite label must apply the same state transition to every record compatible with that label. This creates a structural constraint absent from static erasure coding: each initial color class forces simultaneous successors under every legal assignment. The resulting minimization concerns the *number of possible persistent labels*, even when this number cannot reduce the required binary word length. The distinction is central to the new exact results.

The principal conclusion is the two-level formula in Theorem 11.1. Its ingredients are an exact high-erasure matching normal form (Theorem 4.1), a variational reduction over complement-invariant source sets (Theorem 6.1), an alphabet lower bound for the next erasure layer (Theorem 9.1), and an all-length Hamming-prefix cubical construction (Lemma 7.2). The seven-coordinate exception uses a finite, independently checked propositional refutation supported by source-to-CNF reduction arguments (Theorem 10.2). Section 13 gives the proof's trust boundary and reproduction instructions.

The work follows the characteristic-graph viewpoint in static zero-error coding [1] and uses classical Hamming codes [2, 6]. The alphabet-size measure and blind transition interface differ from the communication objective in oblivious updates [3]. No claim of a general complexity classification for arbitrary horizons is made.

2 Operational model and memory accounting

We use a finite-horizon state model whose static case is zero-error coding with decoder side information [1]. Its transitions receive only the current state and the assigned coordinate–value pair.

Write $[n] = \{1, \dots, n\}$ and let the source be $x \in \mathbb{F}_2^n$. An update $u = (i, b)$ produces

$$U_{i,b}(x)_j = \begin{cases} b, & j = i, \\ x_j, & j \neq i. \end{cases}$$

The memory at the beginning is $M_0 = f(x)$, where $f : \mathbb{F}_2^n \to [K]$. Its online transitions obey

$$M_{t+1} = \Phi_t(M_t, i_t, b_t), \qquad 0 \le t < T. \tag{1}$$

No transition may inspect x_t, the former bit x_{t,i_t}, or uncharged persistent work space. The encoder, transition maps and final decoder are deterministic and may be nonlinear and nonuniform in n. The round index t is public at decoding time; the update sequence remains private.

After any $0 \le t \le T$, the adversary selects $E \subseteq [n]$, $|E| \le q$. The decoder is supplied the current $x_{t,[n]\setminus E}$, together with (M_t, E, t), and must reproduce the full tuple $x_{t,E}$. Every initial vector, legal update sequence, phase and erasure choice must be correct. The requirement also applies when updates reassign the current value or revisit an already modified coordinate.

Definition 2.1. *$R_T^{\mathrm{on}}(n, q)$ is the smallest integer $R \ge 0$ admitting an online state machine with $K \le 2^R$ labels and the above universal recovery guarantee. We assume $n \ge 1$, $T \ge 0$, and $1 \le q \le n$.*

Define $\kappa_T(n, q)$ as the minimum number of persistent memory labels in a valid T-write, q-erasure protocol. Because every bit budget R permits at most 2^R labels, the two resource measures obey $R_T^{\mathrm{on}}(n, q) = \lceil \log_2 \kappa_T(n, q) \rceil$. The integer-valued parameter is used in Section 6.

Any overwrite dictionary forms part of the charged state M_t. In particular, the decoder receives the erasure observation and the phase, together with the stored label, while the update transcript remains private. All state counts refer to alphabet cardinality; explicit combinatorial ranking implements the stated bit budgets.

We set throughout

$$s = T + q, \qquad V(n, r) = \sum_{j=0}^{\min(r,n)} \binom{n}{j}, \qquad D_T(n) = \sum_{k=0}^{\min(T,n)} \binom{n}{k} 2^k.$$

The quantity $D_T(n)$ counts all partial Boolean maps from at most T coordinates to their most recent assigned values.

3 The initial collision restriction

We first strengthen the initial-state collision argument for one erased coordinate to an arbitrary erased set.

Lemma 3.1 (Erasing a collision)**.** *If distinct $x, y \in \mathbb{F}_2^n$ satisfy $f(x) = f(y)$ and $d_{\mathrm{H}}(x, y) \le T + q$, universal zero-error recovery is impossible.*

Proof. Let $D = \{i : x_i \neq y_i\}$, with $d = |D| \ge 1$. If $d \le q$, apply no updates and erase an $E \supseteq D$ of size at most q. The decoder sees equal memory and equal complementary coordinates, yet different erased tuples.

If $d > q$, choose $d - q$ coordinates from D, and apply the same overwrite $U_{i,0}$ in both worlds at those coordinates. This takes at most T operations. Their memory states remain equal by

determinism of each Φ_t, and their current vectors now differ at exactly the q remaining coordinates of D. Erasing those coordinates creates the same decoder observation and contradictory required outputs. The contradiction applies to arbitrary deterministic encoding and transition maps. □

4 The exact matching normal form

Write $X = \mathbb{F}_2^n$ and $N = |X| = 2^n$. For an integer d, let $H_n^{\geq d}$ be the graph on X joining pairs of distinct words with Hamming distance at least d, and write $\nu(H)$ for the cardinality of a maximum matching in a graph H. A matching means a set of vertex-disjoint unordered pairs.

Theorem 4.1 (One-write high-erasure matching principle)**.** *Let $n \geq 3$ and $q \geq 1$ with $3(q+1) > 2n$. Define $G_0 = H_n^{\geq q+2}$ and $G_1 = H_n^{\geq q+1}$. For any matching $M \subseteq E(G_0)$, define*

$$\mathcal{F}(M) = \bigcup_{\{x,y\} \in M} \{\{U_{i,b}(x), U_{i,b}(y)\} : i \in [n],\ b \in \mathbb{F}_2\},$$

where the braces on the right are unordered pairs, with repeated pairs counted once. Let

$$W(M) = \bigcup_{e \in \mathcal{F}(M)} e, \qquad R(M) = X \setminus W(M).$$

Then the unrestricted deterministic optimum is exactly

$$\boxed{\kappa_1(n,q) = \min_{M \text{ matching in } G_0} \max\{N - |M|,\ |\mathcal{F}(M)| + |R(M)| - \nu(G_1[R(M)])\}.} \tag{2}$$

For each candidate M, a protocol attaining the displayed maximum can be constructed using only maximum matching and elementary state lookups. In particular, there is an optimal protocol whose initial fibers are pairs and singletons and whose terminal fibers are exactly the forced edges, a maximum matching of residual vertices, and residual singletons.

Proof. For any three binary words, the sum of their three pairwise Hamming distances is at most $2n$: each coordinate contributes either zero or two. Hence $3(q+1) > 2n$ implies every terminal code with minimum distance $q+1$ has size at most two. The stronger initial minimum distance $q+2$ gives the same conclusion for initial fibers. Thus every initial encoder determines a matching M in G_0; its remaining nonempty initial fibers are singleton words, and $N - |M|$ initial labels are compulsory.

Fix an initial pair $\{x, y\} \in M$ and an update event (i, b). Its two updated words must receive the same terminal label, as the transition acts on their common initial label and identical event. They remain distinct and have distance at least $q+1$, so they constitute a forced pair in G_1. Every *distinct* forced pair demands a distinct terminal label: no valid terminal label can contain three words. Every vertex in the residual set $R(M)$ is reachable at phase one (initialize that word and overwrite one coordinate with its already present value). It cannot be covered by a label reserved for a forced pair. Of the residual vertices, a terminal label covers at most two, and a doubleton must be an edge in $G_1[R(M)]$. Hence at most $\nu(G_1[R(M)])$ disjoint residual doubletons can be formed, forcing at least $|R(M)| - \nu(G_1[R(M)])$ additional terminal labels. This proves the second lower bound.

For achievability, give each source pair in M one initial label, and each remaining initial word its own label. At phase one, assign one label to each distinct forced pair, one label to every edge in a chosen maximum matching of $G_1[R(M)]$, and singleton labels to unmatched residual words. These sets cover X. For an initial paired label, the specified write determines its forced-pair successor label. For an initial singleton label, the write determines a unique current word, and a canonical terminal label containing it can be selected. Both transitions depend only on the stored initial label and (i, b). All fibers respect their corresponding Hamming-distance requirements,

giving universal zero-error recovery at both phases. Phase-specific label names may be reused, so the common cardinality is the maximum of the two phase budgets. Minimizing over M proves (2). □

Remark 4.2 (Exact reduction, not a closed-form optimization)**.** *Formula (2) removes transition maps, encoders and decoders from the outer search. It leaves a finite minimization over matchings in an explicitly defined graph, with one polynomial-time maximum-matching computation for each candidate. Thus it is a strict structural reduction; it does not imply a polynomial-time algorithm in the binary-record length n, nor determine the original general (T, q) complexity class.*

5 Sharp leading constant for the one-write state deficit

We examine the integer state alphabet at the largest erasure budget that permits an initial collision. We now determine its exact first-order asymptotics over *all* blocklengths. The argument has two complementary parts. An exact accounting identity for forced terminal pairs gives a stronger converse, while a family of antipodally symmetric unions of subcubes makes the converse asymptotically tight. Both arguments apply to unrestricted, possibly nonlinear state machines under the operational model of Section 2.

Throughout this section $n \geq 4$, $q = n - 2$, $N = 2^n$, and

$$\Delta_n = N - \kappa_1(n, n-2).$$

A phase-zero fiber containing two source vectors must be an antipodal pair by Lemma 3.1. Let $\mathcal{P}$ denote the collection of those fibers, put $a = |\mathcal{P}|$, and set

$$C = \bigcup_{P \in \mathcal{P}} P, \qquad |C| = 2a.$$

The set C is complement-invariant. Write $e(C)$ for the number of ordinary distance-one edges of the binary cube induced by C, and write $B_1(C) = \{y : d_{\mathrm{H}}(y, C) \leq 1\}$ for its closed radius-one neighborhood.

5.1 An edge–neighborhood accounting identity

Lemma 5.1 (Forced-edge multiplicity and support)**.** *For the source pairs $\mathcal{P}$ above, let $\mathcal{F}$ be the set of distinct unordered terminal pairs compelled by all possible one-coordinate assignments. Then*

$$\boxed{|\mathcal{F}| = 2na - e(C), \qquad \bigcup_{F \in \mathcal{F}} F = B_1(C).} \tag{3}$$

Moreover,

$$\boxed{|B_1(C)| \leq 2(n+1)a - 2e(C).} \tag{4}$$

Proof. Each antipodal source pair $P = \{x, \bar{x}\}$ produces $2n$ distinct output pairs

$$F(P, i, b) = \{U_{i,b}(x), U_{i,b}(\bar{x})\}, \qquad i \in [n], \quad b \in \mathbb{F}_2.$$

The two members agree exactly at coordinate i, with common value b, so that the output pair uniquely identifies (i, b). Every output pair has precisely two possible antipodal predecessor fibers: they are obtained by assigning the two opposite original bits at coordinate i. These two predecessor fibers are simultaneously present in $\mathcal{P}$ precisely when their members form two distance-one edges inside C. Each pair of adjacent antipodal fibers contributes two such cube edges and two duplicated output pairs. Hence the $2na$ generated output pairs lose exactly $e(C)$ occurrences under deduplication, proving the first identity.

The outputs associated with a single source pair are exactly the words at Hamming distance zero or one from either of its members. Taking their union over $\mathcal{P}$ proves the support identity.

Finally, count the incidences (x, y) with $x \in C$ and $y \in B_1(x)$. There are $(n+1)|C| = 2(n+1)a$ incidences. Each $y \in C$ is counted once for its own center and $\deg_C(y)$ additional times for its neighbors in C. Thus at least $\sum_{y \in C} \deg_C(y) = 2e(C)$ of these incidences are repetitions. Every additional intersection of radius-one balls only decreases the union size, proving (4). □

Theorem 5.2 (Sharp universal finite-length converse)**.** *For every integer $n \geq 4$,*

$$\boxed{\kappa_1(n, n-2) \geq 2^n - \left\lfloor \frac{2^n}{2n} \right\rfloor, \qquad \Delta_n \leq \left\lfloor \frac{2^n}{2n} \right\rfloor.} \tag{5}$$

This bound is valid for arbitrary deterministic initialization, nonlinear updates, and phase-dependent zero-error decoders.

Proof. There are $N - a$ different phase-zero labels. At phase one, every member of $\mathcal{F}$ must occupy its own label: two distinct pairs cannot occupy the same label, as each is a pair of current words at distance $n - 1$, and $A(n, n-1) = 2$ for $n \geq 4$. Every current word outside $B_1(C)$ remains reachable after one write by starting from that word and assigning an existing coordinate value. Such a word cannot join a label occupied by a forced pair. The remaining terminal labels support at most two words each. Therefore Lemma 5.1 yields

$$K \geq |\mathcal{F}| + \frac{N - |B_1(C)|}{2} \tag{6}$$

$$\geq 2na - e(C) + \frac{N - 2(n+1)a + 2e(C)}{2} \tag{7}$$

$$= \frac{N}{2} + (n-1)a. \tag{8}$$

Together with the phase-zero count, every admissible protocol satisfies

$$\boxed{K \geq \max\left\{N - a, \frac{N}{2} + (n-1)a\right\}.} \tag{9}$$

The decreasing first affine function and increasing second affine function meet at $a = N/(2n)$, where the common value is $N - N/(2n)$. Taking the integer ceiling of $K \geq N - N/(2n)$ gives (5). □

Forced-edge multiplicity and reachable-neighborhood size cancel the induced-edge contribution $e(C)$ in (9). It also gives $\kappa_1(4, 2) \geq 14$; the matching upper bound follows from two antipodal source pairs.

5.2 Subcube unions that attain the leading constant

Theorem 5.3 (Subcube-union protocol)**.** *Let $n > m \geq 5$, and let $A \subseteq \mathbb{F}_2^m$ be invariant under bitwise complementation and have minimum Hamming distance at least three. Put $t = |A|$ and $a = t2^{n-m-1}$. If $t \leq 2^m/n$, then there is a deterministic one-write, $(n-2)$-erasure protocol with precisely*

$$\boxed{K_0 = N - a, \qquad K_1 = \frac{N}{2} + (n-1)a \leq K_0.} \tag{10}$$

In particular, $\Delta_n \geq a$.

Proof. Write each source word as (u, v) with $u \in \mathbb{F}_2^m$ and $v \in \mathbb{F}_2^{n-m}$. Select the source-pair centers

$$C = A \times \mathbb{F}_2^{n-m}.$$

This set is complement-invariant because A is, and $|C| = t2^{n-m} = 2a$. Give each complementary pair in C one initial label and give every source outside C a singleton label. This requires $K_0 = N - a$ labels.

Put $p = n - m$. There are no distance-one edges between distinct prefix slices indexed by A, because the minimum distance in A is at least three. Every slice is a full p-dimensional subcube, so the induced cube-edge count is

$$e(C) = \frac{p|C|}{2} = pa.$$

The disjoint radius-one neighborhoods of the prefix codewords yield

$$B_1(C) = B_1(A) \times \mathbb{F}_2^p, \qquad |B_1(C)| = t(m+1)2^p = 2a(m+1).$$

For each initial pair and write event, allocate the forced output pair as one terminal fiber; deduplicate pairs common to two source pairs. By Lemma 5.1, this creates $2na - pa = a(n+m)$ terminal fibers. The words not covered by these pairs form a complement-invariant set, since both C and its radius-one neighborhood are complement-invariant. Partition this remaining set into complementary pairs; each is a valid terminal fiber of distance n. Every forced fiber has distance $n-1$. Thus all phase-one fibers admit unique recovery of up to $n-2$ erased coordinates, and their total number is

$$K_1 = a(n+m) + \frac{N - 2a(m+1)}{2} = \frac{N}{2} + (n-1)a.$$

An initial paired fiber transitions to its forced output pair under each assignment. An initial singleton transitions to any terminal fiber containing its updated word. Both choices depend only on the initial label and the write event. The condition $a \le N/(2n)$ is equivalent to $t \le 2^m/n$ and guarantees $K_1 \le K_0$, proving (10). □

Lemma 5.4 (Polynomial-size prefix selection)**.** *Let $\ell = \lceil \log_2 n \rceil$ and $m = \lceil \log_2(n\ell^2) \rceil$. For every $n \ge 1024$ there is a deterministically computable complement-invariant code $A \subseteq \mathbb{F}_2^m$ of minimum distance three and cardinality*

$$t = 2\left\lfloor \frac{2^m}{2n} \right\rfloor \ge \frac{2^m}{n} - 2. \tag{11}$$

The prefix code can be found by a greedy procedure in time polynomial in n.

Proof. Write $V(m,2) = 1 + m + \binom{m}{2}$. The space $\mathbb{F}_2^m$ contains 2^{m-1} complementary pairs. Greedily choose pairs $\{u, \bar{u}\}$ while requiring their members to be at distance at least three from all previously selected members. Selecting one pair eliminates at most $2V(m,2)$ candidate pairs, since each of its two members has at most $V(m,2)$ words at distance at most two. The procedure can therefore select at least $2^{m-1}/[2V(m,2)]$ complementary pairs. The required number, $t/2 = \lfloor 2^{m-1}/n \rfloor$, is attainable whenever $n \ge 2V(m,2)$.

For $n \ge 1024$, the elementary inequalities $\ell^2 \le 2^{\ell-1}$ and $n \ge 4\ell^2$ imply

$$m \le 2\ell - 1, \qquad 2V(m,2) = m^2 + m + 2 \le (m+1)^2 \le 4\ell^2 \le n.$$

The greedy criterion is thus satisfied. Since $2^m < 2n\ell^2$, exhaustive enumeration of all 2^{m-1} complementary pairs and comparison with the selected pairs takes polynomial time in n. The choice automatically gives complement-invariance and the stated minimum distance. □

Theorem 5.5 (Sharp first-order state-alphabet asymptotics)**.** *For every integer $n \ge 1024$, with $\ell = \lceil \log_2 n \rceil$,*

$$\boxed{\left(1 - \frac{2}{\ell^2}\right)\frac{2^n}{2n} \le \Delta_n \le \left\lfloor \frac{2^n}{2n} \right\rfloor.} \tag{12}$$

Consequently, along all *positive blocklengths,*

$$\boxed{\lim_{n\to\infty} \frac{2n\,[2^n - \kappa_1(n, n-2)]}{2^n} = 1.} \tag{13}$$

The lower bound is realized by a deterministic construction based on complement-invariant prefix codes and subcube unions.

Proof. Apply Lemma 5.4 to Theorem 5.3. The resulting source-pair count is

$$a = \frac{t2^{n-m}}{2} \geq \frac{2^n}{2n} - \frac{2^n}{2^m} \geq \left(1 - \frac{2}{\ell^2}\right)\frac{2^n}{2n},$$

using $2^m \geq n\ell^2$. Since $t \leq 2^m/n$, the constructed alphabet contains $N - a$ labels, and therefore $\Delta_n \geq a$. The universal upper bound is Theorem 5.2. The two normalized bounds converge to one and prove (13). □

6 An exact variational principle and finite optima

We sharpen the general converse by an exact finite combinatorial formulation. This formulation eliminates arbitrary transition maps from the optimization, reduces the state-alphabet problem to overlap statistics of antipodal subsets of the Boolean cube, and produces an infinite family of exact finite optima.

Write $\bar{x} = x \oplus \mathbf{1}$ and $\overline{C} = \{\bar{x} : x \in C\}$. For any complement-invariant subset $C = \overline{C} \subseteq \mathbb{F}_2^n$, write $a = |C|/2$ and

$$d_C(y) = |\{i \in [n] : y \oplus e_i \in C\}|, \qquad \Xi(C) = \sum_{y \notin C} (d_C(y) - 1)_+, \qquad (u)_+ = \max\{0, u\}.$$

The integer $\Xi(C)$ counts the excess of overlapping radius-one neighborhoods at words outside C. Complementation preserves d_C, so $\Xi(C)$ is even.

Theorem 6.1 (Exact finite state-alphabet variational principle)**.** *For every $n \geq 4$, let $N = 2^n$. Then the minimum state-alphabet cardinality for one blind overwrite with up to $n-2$ erasures equals*

$$\boxed{\kappa_1(n, n-2) = \min_{\substack{C \subseteq \mathbb{F}_2^n \\ C = \overline{C}}} \max\left\{N - \frac{|C|}{2},\quad \frac{N}{2} + \frac{(n-1)|C|}{2} + \frac{\Xi(C)}{2}\right\}.} \tag{14}$$

Every set C in this minimization induces an explicit deterministic protocol achieving the displayed maximum, with only one- or two-element initial and terminal state supports.

Proof. Fix any valid protocol. By the initial collision lemma with $T+q = n-1$, each non-singleton initial state fiber is an antipodal pair. Let C be the union of these pairs, and put $a = |C|/2$. At phase zero, exactly $N - a$ nonempty initial labels are required.

Let $e(C)$ denote the induced distance-one edge count. Lemma 5.1 identifies exactly $2na - e(C)$ distinct forced phase-one pairs, with union of endpoints $B_1(C)$. Each forced pair must use its own label because the required phase-one minimum distance is $n-1$ and $A(n, n-1) = 2$. No such label may additionally contain a word outside $B_1(C)$. At most two words can share any other phase-one label.

Counting the radius-one ball incidences more precisely gives an *identity*:

$$|B_1(C)| = 2(n+1)a - 2e(C) - \Xi(C). \tag{15}$$

Indeed, a point $y \in C$ appears once as its own center and $d_C(y)$ more times among the balls of its neighbors, accounting for $\sum_{y \in C} d_C(y) = 2e(C)$ repeated incidences. A point $y \notin C$ occurs

$d_C(y)$ times and contributes $(d_C(y)-1)_+$ repeats. Summing these disjoint kinds of repetitions yields (15). It follows that the phase-one alphabet requires at least

$$\begin{aligned} K_1 &\geq (2na - e(C)) + \frac{N - |B_1(C)|}{2} \\ &= \frac{N}{2} + (n-1)a + \frac{\Xi(C)}{2}. \end{aligned}$$

Taking the larger of the phase-zero and phase-one lower bounds proves that every valid protocol uses at least the right side of (14), evaluated at its own C.

Conversely, fix any complement-invariant C. Use its a antipodal pairs as phase-zero doubleton labels and all remaining words as phase-zero singleton labels. Give each distinct forced phase-one pair its own terminal label. The residual words $\mathbb{F}_2^n \setminus B_1(C)$ are complement-invariant; partition them into antipodal pairs, each receiving one further terminal label. Their count is exactly $(N - |B_1(C)|)/2$, so equality holds in the displayed terminal count. Map each paired initial label to the forced output pair specified by the write event, and each singleton initial label to a canonical terminal label containing its updated word. The transition requires only the current label and the write event. Initial doubletons have Hamming distance n and terminal doubletons have distance $n-1$ or n; therefore every set of at most $n-2$ erased bits is uniquely reconstructible at both phases. Public phase information permits reuse of the larger of the two label alphabets. This achieves the maximum in (14), and minimization over C completes the proof. □

Theorem 6.2 (Exact optimal alphabets at all dyadic blocklengths)**.** *For every integer $r \geq 2$ and $n = 2^r$,*

$$\boxed{\kappa_1(n, n-2) = 2^n - \frac{2^{n-1}}{n}, \qquad \Delta_n = \frac{2^{n-1}}{n}.} \tag{16}$$

These integer optima are attained by explicitly indexed, polynomial-time computable state machines.

Proof. Theorem 5.2 gives the lower bound $\kappa_1(n, n-2) \geq N - N/(2n)$, with no rounding because n is a power of two.

Put $m = r + 1$ and $p = n - m \geq 1$. Set

$$A = \{0^m, 1^m\}, \qquad C = A \times \mathbb{F}_2^p.$$

The two prefix words have distance $m \geq 3$, and C is closed under complementation. It consists of 2^{p+1} vectors, hence $a = 2^p = 2^{n-r-1} = N/(2n)$ antipodal pairs. A word outside C has at most one adjacent word in C: changing a suffix coordinate preserves its excluded prefix, whereas changing one prefix coordinate could enter at most one of the two antipodal prefix values because $m \geq 3$. Therefore $\Xi(C) = 0$.

The variational principle supplies an online protocol with

$$K_0 = N - a, \qquad K_1 = N/2 + (n-1)a.$$

Substituting $a = N/(2n)$ gives $K_1 = N - a = K_0$. Thus the universal lower bound is attained. The structure of C permits direct ranking of initial antipodal fibers and terminal forced edges using their prefix, suffix and overwrite coordinates by explicitly ranking the prefix and suffix indices; the $m = 3, 4$ cases are handled by the same explicit formulas. These rankings, transition maps and decoder procedures run in polynomial time in n. □

Proposition 6.3 (The first non-dyadic exact optimum)**.** *For five coordinates, one blind overwrite, and up to three erased coordinates,*

$$\boxed{\kappa_1(5,3) = 29, \qquad R_1^{\mathrm{on}}(5,3) = 5.}$$

Proof. Theorem 5.2 gives $\kappa_1(5,3) \geq 32 - \lfloor 32/10 \rfloor = 29$. For the matching construction, let C consist of the three antipodal pairs

$$\{00000, 11111\}, \qquad \{00001, 11110\}, \qquad \{00010, 11101\}.$$

Then $a = 3$ and direct adjacency counting gives $\Xi(C) = 2$. Formula (14) yields

$$K = \max\{32 - 3,\ 16 + 4 \cdot 3 + 2/2\} = 29.$$

The variational proof constructs the required update and decoder maps. □

Corollary 6.4 (Structure of exact dyadic extremizers)**.** *Let $n = 2^r \geq 4$. For a complement-invariant initial paired-source set C, the canonical construction in Theorem 6.1 uses the optimum $\kappa_1(n, n-2)$ if and only if*

$$\boxed{|C| = 2^n/n, \qquad \Xi(C) = 0.}$$

Equivalently, each current word outside C has at most one distance-one neighbor in C.

Proof. Write $a = |C|/2$ and $N = 2^n$. Equality in the lower bound $K \geq \max\{N - a, N/2 + (n-1)a + \Xi/2\}$ at $K = N - N/(2n)$ forces $a \geq N/(2n)$ from the first term and $a \leq N/(2n) - \Xi/[2(n-1)]$ from the second. Hence $a = N/(2n)$ and $\Xi = 0$. The reverse implication follows by substitution in (14). The last equivalence is immediate from the nonnegative summands in the definition of $\Xi(C)$. □

n	$q = n - 2$	$\kappa_1(n, q)$	$2^n - \kappa_1$	certificate
4	2	14	2	dyadic exact
5	3	29	3	variational exact
8	6	240	16	dyadic exact
16	14	63488	2048	dyadic exact

Table 1: New exact integer state counts. The finite-size verifier enumerates every antipodal initial-pair selection for $n = 4, 5$, and checks complete transition/erasure observations for explicit optima at $n = 4, 5, 8$.

The finite optimization (14) also provides a direct route to optimality certificates and a natural combinatorial problem: minimize the exterior overlap functional $\Xi(C)$ over complementary-pair subsets of the Boolean cube. The dyadic extremizers combine a precise source-pair density with collision-free exterior neighborhoods. Other blocklengths admit an integer tradeoff between source-pair density and exterior overlap; the length-five optimum is the smallest instance in which positive overlap appears in an optimal protocol. The accompanying `verification/verify_folded_cube_exception.py` proves the seven-dimensional obstruction by exhaustive closure search, and `verification/independent_v10_audit.py` separately enumerates all admissible source-pair comparisons at $n = 7, 8$. The V12 protocol checker independently enumerates the seven-dimensional write and erasure observations.

7 The complete finite-blocklength solution of the one-write alphabet problem

The variational formula (14) admits a complete finite-blocklength solution. The remaining ingredient is a mixed-size subcube selection inside a perfect binary Hamming code. The seven-dimensional exceptional value is supported by two complementary finite approaches: an exact

combinatorial closure search and an machine-checked DRUP certificate for the adjacent erasure layer.

Throughout this section $n \geq 4$, $N = 2^n$, $M = 2^{n-1}$, and

$$D_n = N - \kappa_1(n, n-2), \qquad h_n = \left\lfloor \frac{M}{n} \right\rfloor.$$

Theorem 5.2 gives the universal inequality $D_n \leq h_n$.

Theorem 7.1 (Exact optimum for every blocklength)**.** *For every $n \geq 4$, the unrestricted deterministic one-write alphabet optimum satisfies*

$$\boxed{\kappa_1(n, n-2) = 2^n - \left\lfloor \frac{2^{n-1}}{n} \right\rfloor + \mathbf{1}_{\{n=7\}}.} \tag{17}$$

All instances admit explicit finite deterministic online protocols achieving the stated state cardinality. For every $n \geq 8$, an optimal initial source-pair set can be chosen to have zero exterior neighborhood overlap, $\Xi(C) = 0$.

Lemma 7.2 (Hamming-prefix cubical packing)**.** *For every $n \geq 8$ there exists a complement-invariant set $C \subseteq \mathbb{F}_2^n$ such that*

$$\boxed{|C| = 2h_n, \qquad \Xi(C) = 0.}$$

It can be described explicitly by a perfect binary Hamming code of length $2^r - 1$, where $r = \lfloor \log_2 n \rfloor$.

Proof. Write $d = 2^r$ and $m = d - 1$; thus $d \leq n < 2d$ and $p = n - m = n - d + 1 \in [1, d]$. Let $H \subseteq \mathbb{F}_2^m$ be the binary Hamming code whose parity-check columns are precisely the nonzero vectors of $\mathbb{F}_2^r$. Its minimum Hamming distance is three. The sum of the check columns is zero for $r \geq 3$, so $\mathbf{1} \in H$ and H partitions into

$$B = 2^{m-r-1} = 2^{d-r-2}$$

complementary pairs $\{u, \bar{u}\}$.

For each of these B pairs, choose either no suffix points or an affine subcube $D_i \subseteq \mathbb{F}_2^p$ of cardinality 2^{j_i}, $0 \leq j_i \leq p$. Add the two sets $\{u_i\} \times D_i$ and $\{\bar{u}_i\} \times \overline{D_i}$ to C. This produces a complement-invariant union of subcubes. Different prefix codewords are at distance at least three; each suffix subset is a subcube. Consequently, an exterior point has at most one neighbor in C, yielding $\Xi(C) = 0$. The number of complementary source pairs is $a = \sum_i |D_i|$.

We show that $a = h_n$ can be realized using at most B suffix subcubes, apart from one explicitly handled length. Put $L = 2^p$ and write $h_n = qL + t$, with $0 \leq t < L$. Selecting q full suffix cubes and one cube of size 2^j for each 1-bit of t realizes h_n using $q + \mathrm{popcount}(t)$ of the B prefix pairs. Since

$$BL = 2^{n-r-1} = \frac{M}{d}, \qquad h_n \leq \frac{M}{n},$$

we have $q \leq Bd/n$. If $n = d$, then $h_n = BL$, $q = B$, and $t = 0$. Otherwise set $u = n - d \in [1, d-1]$, so $p = u + 1$. For $r \geq 4$ we have $B = 2^{d-r-2} \geq 2(d+1)$, and hence

$$B - q \geq \frac{Bu}{d+u} \geq u + 1 = p \geq \mathrm{popcount}(t),$$

where the middle step follows from

$$2(d+1)u - (d+u)(u+1) = (u-1)(d-u) \geq 0.$$

Thus the chosen number of prefix pairs never exceeds B.

For $r = 3$, we have $d = 8$, $m = 7$, $B = 8$, and $8 \le n \le 15$. The values $(n, h_n, q + \mathrm{popcount}(t))$ are

$$(8, 16, 8), (9, 28, 7), (10, 51, 8), (11, 93, 8), (12, 170, 7), (13, 315, 9), (14, 585, 7), (15, 1092, 6).$$

Only $n = 13$ uses nine pieces with eight prefix pairs. Here $p = 6$ and $315 = 4{\cdot}64{+}32{+}16{+}8{+}2{+}1$. Place the final size-two cube $\{000000, 000001\}$ and the singleton $\{111100\}$ under the same remaining prefix pair, keeping the other seven occupied prefix pairs unchanged. These two suffix components have distance at least four, so the resulting C still satisfies $\Xi(C) = 0$, and its pair count is 315. Every length is covered. □

Proof of Theorem 7.1. For $n \ge 8$, choose C from Lemma 7.2, giving $a = h_n$ and $\Xi(C) = 0$. The exact variational principle (14) constructs a valid protocol with

$$K = \max\{N - h_n, M + (n-1)h_n\} = N - h_n,$$

because $nh_n \le M$. The universal converse $D_n \le h_n$ proves equality.

At $n = 4$, the two initial antipodal pairs with representatives $0, 1$ have $a = 2$ and $\Xi(C) = 0$, giving $\kappa_1(4, 2) = 14$. At $n = 5$, the representatives $0, 1, 2$ give $a = 3$, $\Xi(C) = 2$, and $\kappa_1(5, 3) = 29$, recovering Proposition 6.3. At $n = 6$, take the five representatives $5, 10, 21, 23, 26$ (with all words interpreted as six-bit vectors) and their antipodes. Direct neighborhood counting gives $a = 5$, $\Xi(C) = 4$, and $\kappa_1(6, 4) = 59$. In each case, the universal converse coincides with the construction.

For $n = 7$, the converse gives $\kappa_1(7, 5) \ge 119$. The length-seven perfect Hamming code yields $a = 8$, $\Xi(C) = 0$, hence $\kappa_1(7, 5) \le 120$. Lemma 7.3 below rules out 119, proving $\kappa_1(7, 5) = 120$. This completes the proof for all $n \ge 4$. □

Lemma 7.3 (The unique seven-coordinate exception)**.** *In the seven-dimensional cube, every complement-invariant set of size eighteen has $\Xi(C) \ge 6$. Consequently $\kappa_1(7, 5) = 120$.*

Proof. We prove the stronger statement that $a = 9$ forces $\Xi(C)/2 \ge 3$. Identify antipodal pairs in Q_7 to obtain the folded six-cube F with 64 vertices $0, \dots, 63$ and neighbor sets

$$\mathcal{N}(v) = \{v \oplus 2^i : 0 \le i < 6\} \cup \{v \oplus 63\}.$$

A complement-invariant C corresponds to a nine-vertex set $S \subset F$, and $\Xi(C)/2 = \sum_{v \notin S} (|\mathcal{N}(v) \cap S| - 1)_+$.

For a set E of at most two exterior exceptional vertices, define the closure operator $\mathrm{cl}_E(A)$ by repeatedly adjoining each $v \notin A \cup E$ that has at least two neighbors already in A. This is a monotone closure: if a target S has no repeated exterior neighbors outside E, then $\mathrm{cl}_E(A) \subseteq S$ for every $A \subseteq S$. The search adds one allowed vertex at a time, closes the new set, and prunes any closure exceeding nine vertices or containing a forbidden vertex. It explores every resulting closed set, with repeated sets memoized. This recursion is exhaustive: if a valid closed nine-vertex target S contains an already explored closed A, choosing any $v \in S \setminus A$ yields a closure contained in S; iteration must reach S.

For $\Xi(C)/2 = 0$, graph translations let us assume $0 \in S$. An exact closed-set search enumerates 7,549 closed sets and 446,992 extensions without finding a valid nine-vertex target. For $\Xi(C)/2 = 1$, exactly one exterior vertex has two selected neighbors. Translate it to 0 and use coordinate permutations of the seven-dimensional cube to normalize those neighbors as $1, 2$. Forbid the other five neighbors of 0. This search enumerates 26 closed sets and 1,390 extensions, again without a target.

Finally $\Xi(C)/2 = 2$ has precisely two possibilities: one exceptional exterior vertex with three selected neighbors, or two exceptional vertices each with two selected neighbors. In the first case translate the exception to 0 and enumerate the $\binom{7}{3} = 35$ choices of its selected neighbors. In the second, vertex transitivity and distance transitivity of the folded cube permit the two exceptions to

be represented by 0 and one of $1, 3, 7$, corresponding to distances one, two, and three. For each such position, enumerate all $\binom{7}{2}^2 = 441$ choices of their selected-neighbor pairs. In every instance, the remaining neighbors of an exceptional vertex are forbidden and the closure search proceeds exactly as above. The independent Python checker `verification/verify_folded_cube_exception.py` exhausts these $35 + 3 \cdot 441 = 1,358$ cases and finds no target, visiting 1,721 closed states over those cases. The same script checks the zero-overlap case (7,549 states; 446,992 extensions) and the one-overlap case (26 states; 1,390 extensions), without calling a SAT solver or importing the V12 CNF implementation. All computations consist of 64-bit masks, bit counts, and finite exhaustive recursion with the monotonicity justification given here. The verified script, its raw output, and the source code are archived with the manuscript. It follows that $\Xi(C)/2 \geq 3$ for every nine-pair set C. This lower bound is attained by the nine representative source words $8, 12, 19, 23, 40, 44, 51, 55, 56$ together with their seven-bit complements: direct integer adjacency counting gives $\Xi(C) = 6$.

Finally, if a valid 119-label protocol existed, its paired-source set would need $a \geq 9$ from the initial label count. The terminal lower bound $K \geq 64 + 6a + \Xi(C)/2$ forces $a \leq 9$, hence $a = 9$ and $\Xi(C)/2 \leq 1$, contrary to the certificate. Thus $K \geq 120$ and the Hamming construction is optimal. □

Corollary 7.4 (An exact all-length optimality criterion)**.** *Let $\delta_n = h_n - \mathbf{1}_{\{n=7\}}$, and let $C = \bar{C}$ be any set of initially paired words. The canonical protocol induced by C is optimal if and only if*

$$\boxed{a \geq \delta_n, \qquad (n-1)a + \Xi(C)/2 \leq M - \delta_n.} \tag{18}$$

For $n \geq 8$ write $M = nh_n + \rho_n$ with $0 \leq \rho_n < n$. In that regime the criterion is equivalent to either

$$a = h_n, \quad \Xi(C)/2 \leq \rho_n,$$

or, when $\rho_n = n - 1$, to $a = h_n + 1$ and $\Xi(C) = 0$. For the exceptional length $n = 7$, every optimal source-pair set has $a = 8$ and $\Xi(C)/2 \leq 8$.

Proof. Substitute the exact optimum $K = N - \delta_n$ of Theorem 7.1 in both affine terms of (14); this gives (18) and proves necessity and sufficiency. For $n \geq 8$, put $a = h_n + t$ with $t \geq 0$ and simplify to $(n-1)t + \Xi(C)/2 \leq \rho_n$. As $\rho_n < n$, only $t = 0$ is possible, except that $t = 1$ can occur if $\rho_n = n - 1$ and $\Xi(C) = 0$. For $n = 7$ the criterion initially permits $a = 8$ or $a = 9$; the exact $\Xi(C)/2 \geq 3$ certificate excludes $a = 9$, yielding the final assertion. □

Corollary 7.5 (Computational complexity of the unrestricted optimum)**.** *The unrestricted state-alphabet optimum $n \mapsto \kappa_1(n, n-2)$ is computable by exact integer arithmetic in polynomial time in the output length $n + 1$. Its threshold decision problem, given n and K in binary, "is $\kappa_1(n, n-2) \leq K$?", belongs to the class $\mathbf{P}$. This holds independently of the complexity of candidate-restricted variants of the cubical overlap problem.*

Proof. For $n \geq 4$, formula (17) uses one shift, integer division, subtraction and the test $n = 7$. The numerical answer has n binary digits. For the threshold language, compare the bit length of K to n: if it is shorter than n, reject, and if longer than $n + 1$, accept; in the remaining cases n is bounded by the input size of K and the exact formula is computable in polynomial time. The finite cases $n < 4$ may be handled separately. □

8 The second near-maximal erasure layer

For this section let $n \geq 7$, $T = 1$, $q = n - 3$, $N = 2^n$. Every initial fiber has minimum distance $n - 1$ and every terminal fiber minimum distance $n - 2$. In this range a code in either phase contains at most two elements, by the three-word distance obstruction. Thus source doubletons form a matching M of a pairs with endpoint set C.

n	h_n	δ_n	$\kappa_1(n, n-2)$	certificate
4	2	2	14	two pairs
5	3	3	29	positive-overlap construction
6	5	5	59	positive-overlap construction
7	9	8	120	exact closed-set certificate
8	16	16	240	cubical Hamming prefix
9	28	28	484	cubical Hamming prefix
10	51	51	973	cubical Hamming prefix
11	93	93	1955	cubical Hamming prefix
12	170	170	3926	cubical Hamming prefix
13	315	315	7877	two-component suffix
16	2048	2048	63488	dyadic Hamming prefix

Table 2: Exact optimal label cardinalities for one blind overwrite with $q = n - 2$. All values use the closed-form theorem, with finite source-set witnesses.

8.1 Forced-pair collisions and neighborhoods

For an initial pair $P = \{x, y\}$, set

$$\mathcal{F}(P) = \{\{U_{i,b}(x), U_{i,b}(y)\} : i \in [n],\ b \in \mathbb{F}_2\}, \qquad \mathcal{F}(M) = \bigcup_{P \in M} \mathcal{F}(P).$$

Each admissible source pair has distance n or $n-1$. Consequently $\mathcal{F}(P)$ has exactly $2n$ distinct edges, with terminal distance $n-1$ or $n-2$. Write $D = 2na - |\mathcal{F}(M)|$ for repeated forced-edge occurrences. Also let $e(C)$ count the unordered edges of the ordinary hypercube with both endpoints in C.

Lemma 8.1 (Local classification of common successor pairs)**.** *Assume $n \geq 7$. Let P, Q be disjoint two-element source sets, each of Hamming distance at least $n-1$, and let $\mathcal{F}(P)$ and $\mathcal{F}(Q)$ denote their respective forced one-write successor sets. If these sets intersect, there are exactly two ordinary cube edges between P and Q. Moreover, if both edges have the same coordinate, the intersection has at most two elements; if their coordinates differ, the intersection has at most one element. No three pairwise disjoint legal source pairs can force a common successor pair.*

Proof. Let $E = \{u, v\}$ be a common successor pair. Its distance is either $n-1$ or $n-2$, because one assignment can eliminate at most one mismatch. Every source pair that produces E has to be obtained by undoing an assignment at a coordinate where u and v agree.

When $d_{\mathrm{H}}(u, v) = n-1$, the agreement coordinate i is unique. The possible source pairs consist of $\{u, v\}$, its translate $\{u \oplus e_i, v \oplus e_i\}$, and the two opposite single-endpoint reversals $\{u \oplus e_i, v\}$ and $\{u, v \oplus e_i\}$. Any subcollection of these pairs with mutually disjoint endpoints has size at most two. Two disjoint members occupy opposite sides of the coordinate-i four-cycle, and its two cross-pair cube edges both have coordinate i.

When $d_{\mathrm{H}}(u, v) = n-2$, write i, j for the two agreement coordinates. A legal source pair giving E must reverse exactly one endpoint at exactly one of i, j. The four candidates are $\{u \oplus e_i, v\}$, $\{u, v \oplus e_i\}$, $\{u \oplus e_j, v\}$ and $\{u, v \oplus e_j\}$. All contain either u or v, so at most two candidates can be pairwise disjoint. For two disjoint candidates, one reverses u and the other reverses v; the two resulting cross-pair unit edges have coordinates i and j, possibly equal.

These cases prove the two-edge necessity and exclude triple-disjoint predecessors. To count the intersection for fixed disjoint P, Q, observe that an ordinary cube edge between their endpoints can only be made into a common output by assigning its changed coordinate. The two necessary cross-pair edges fix the overwritten coordinate on each side. If those coordinates agree, only the two possible assigned values remain, so there are at most two common outputs. If they differ, both final output values are forced by the untouched opposite endpoint, leaving at most one common output. This gives the claimed bounds on $|\mathcal{F}(P) \cap \mathcal{F}(Q)|$. □

Lemma 8.2 (Collision–adjacency inequality)**.** *For $n \geq 7$ and source matching M as above,*

$$\boxed{D \leq e(C), \qquad |\mathcal{F}(M)| \geq 2na - e(C).}$$

Proof. First consider two disjoint source pairs P, Q, each of length at least $n-1$. Lemma 8.1 describes all possible shared successor pairs. Let $e(P,Q)$ be the number of hypercube edges between their endpoints. Each endpoint of P is adjacent to at most one endpoint of Q, by the triangle inequality and $d_{\mathrm{H}}(Q) \geq n-1 > 2$, so $e(P,Q) \leq 2$. By Lemma 8.1, a shared forced edge requires two cross-pair unit edges, and the overlap contains at most two forced edges. Hence

$$|\mathcal{F}(P) \cap \mathcal{F}(Q)| \leq e(P,Q).$$

The same lemma excludes a common forced edge from three pairwise disjoint source pairs. Thus the multiplicity correction contains only pairwise intersections. Consequently

$$D = \sum_{P<Q\in M} |\mathcal{F}(P) \cap \mathcal{F}(Q)| \leq \sum_{P<Q\in M} e(P,Q) \leq e(C).$$

The final inequality also uses that any two words inside an initial source pair are at distance at least $n-1$ and therefore do not form a hypercube edge. □

Write $B_1(C) = \{y \in X : d_{\mathrm{H}}(y,C) \leq 1\}$, and for an exterior word $y \notin C$ let $d_C(y) = |\{x \in C : d_{\mathrm{H}}(x,y) = 1\}|$. Define

$$\Xi(C) = \sum_{y\notin C} (d_C(y) - 1)_+, \qquad (u)_+ = \max\{u, 0\}.$$

No complement invariance of C is assumed. A write to a paired source yields precisely the source words and all their Hamming neighbors, so the endpoint union of the forced terminal pairs is $B_1(C)$.

Lemma 8.3 (Exact union identity)**.** *For any $C \subseteq \mathbb{F}_2^n$ with $|C| = 2a$,*

$$\boxed{|B_1(C)| = 2(n+1)a - 2e(C) - \Xi(C).}$$

Proof. There are $2(n+1)a$ incidences between centers $x \in C$ and points at distance at most one from x. Every $y \in C$ receives one incidence from itself and $\deg_C(y)$ extra incidences, totaling $2e(C)$ center repetitions. An exterior y with $d_C(y) \geq 1$ receives exactly $d_C(y) - 1$ extra incidences. Subtracting both kinds of repetition gives the union cardinality. □

9 A sharp converse and all-length optimum outside dimension seven

Theorem 9.1 (Geometric converse)**.** *For every $n \geq 7$ and $q = n-3$,*

$$\boxed{\kappa_1(n, n-3) \geq N - \left\lfloor \frac{N}{2n} \right\rfloor.}$$

More precisely, every source matching with a pairs and endpoint set C requires

$$K \geq \max\left\{N - a,\ \frac{N}{2} + (n-1)a + \frac{\Xi(C)}{2} + e(C) - D\right\}.$$

Proof. The initial state partition requires $N - a$ labels. Every distinct forced terminal pair needs its own terminal label, since a terminal label may contain at most two words. All N current words are reachable: initialize each word and reassign an existing coordinate bit. Every word outside $B_1(C)$ must use a further label, and one terminal label can serve at most two such residual words. Therefore

$$K \geq |\mathcal{F}(M)| + \frac{N - |B_1(C)|}{2} = \frac{N}{2} + (n-1)a + e(C) - D + \frac{\Xi(C)}{2}.$$

By Lemma 8.2, $e(C) - D \geq 0$; also $\Xi(C) \geq 0$. Thus

$$K \geq \max\{N - a,\ N/2 + (n-1)a\}.$$

The two affine expressions agree at $a = N/(2n)$, and their maximum is at least $N - N/(2n)$ for every real a. Since K is integer, $K \geq N - \lfloor N/(2n) \rfloor$. □

10 The seven-coordinate computer-assisted exclusion

The general converse at $n = 7$, $q = n - 3 = 4$ gives $K \geq 119$. The antipodal length-seven Hamming construction with eight initial paired labels and 112 singleton labels gives $K = 120$. We describe the exact finite obstruction for the remaining possibility, $K \leq 119$, and the independently checked Boolean certificate for excluding it.

Lemma 10.1 (Necessary conditions for 119 labels)**.** *Let M be the initial matching of a pairs of Hamming distance six or seven, and let C be its endpoint set. Write $D = 14a - |\mathcal{F}(M)|$ for duplicated forced-terminal pairs, $e = e(C)$ for induced cube edges, and $\Xi = \Xi(C)$ for exterior Hamming-neighborhood overlap. Any protocol with at most* 119 *labels satisfies*

$$a = 9, \qquad e - D + \left\lceil \frac{\Xi}{2} \right\rceil \leq 1.$$

Proof. The initial phase uses at least $128 - a$ labels. Every selected source pair produces 14 forced terminal pairs and $|B_1(C)| = 16a - 2e - \Xi$. The terminal phase uses at least

$$14a - D + \left\lceil \frac{128 - |B_1(C)|}{2} \right\rceil = 64 + 6a + e - D + \left\lceil \frac{\Xi}{2} \right\rceil.$$

The collision–adjacency lemma gives $D \leq e$. If $K \leq 119$ then the first count gives $a \geq 9$ and the second gives $a \leq 9$; substitution proves the displayed inequality. □

A source pair has two distinct endpoints and distance either six or seven. There are exactly 512 eligible initial pairs. The conditions in Lemma 10.1 split into two alternatives: (B) $\Xi = 0$, and (A) $e - D = 0$ with $\Xi \leq 2$. In case B the cube-isometry group normalizes a first pair to one of the distance-six or antipodal representatives. In case A the neighborhood formula and $|B_1(C)| \leq 128$ imply $e \geq 7$, so at least one strong pair collision occurs. Exhaustive pair-pair enumeration finds precisely three isometry orbits for such collisions, giving five exhaustive representative cases.

Five-case completeness and formalization boundary. A hypothetical 119-label protocol necessarily has $a = 9$ and $e - D + \lceil \Xi/2 \rceil \leq 1$. Since $e - D$ and Ξ are nonnegative integers, either $\Xi = 0$ (case B) or $e = D$ and $\Xi \leq 2$ (case A). In case B, a first source pair can be normalized to $(0, 127)$ if all chosen pairs are antipodal, or to $(0, 63)$ if any pair has distance six. In case A, the closed-ball identity gives $e \geq 7$ and every positive cross-pair adjacency must be offset by a forced-edge collision. As a pair of source pairs with exactly one cross-unit edge has no common successor edge, at least one pair of source pairs has two shared forced edges. The independent orbit enumeration normalizes that pair into exactly three pair-pair isometry types, represented

Case	DIMACS variables	Clauses	Checked RUP additions
B.antipodal	1197	7080	801
B.mixed	10285	44968	32062
A.normal	18217	91673	4860
A.special	18217	91673	5706
A.antipodal	18217	91673	6853
Total			50282

Table 3: Five propositional proof cases for the seven-coordinate exclusion. Each proof is an addition-only DRUP derivation ending in the empty clause; the exact checker validates every added clause by unit propagation.

by $((0,63),(1,62))$, $((0,63),(64,127))$ and $((0,127),(1,126))$. The frozen CNF files separately encode the remaining choices subject to these representative constraints. The correctness of this normalization is a mathematical prerequisite to the Boolean UNSAT proof; it is verified by the accompanying finite orbit enumerator, not inferred from the DRUP checker.

Each CNF uses source-pair selection variables, exact-nine and disjointness constraints, indicators for selected endpoints, and a Boolean encoding of the required external-neighborhood bounds. The five CNFs are regenerated byte-identically by the supplied source. A separate CDCL implementation generated the addition-only DRUP traces. This certificate design follows the clausal-proof verification methodology of Wetzler, Heule and Hunt [7]; an external LRAT or independently verified checker would provide a further assurance layer [8]. The proof checker parses the original CNF, then checks every added clause using the RUP rule: assuming the negations of its literals must force a conflict under exact Boolean unit propagation. An empty clause completes each refutation. The bundled checker reports the counts in Table 3; truncating the proof or introducing an unjustified clause is rejected. The cardinality gates were subjected to 2,046 truth-table and unit-propagation tests.

Theorem 10.2 (Seven-coordinate optimum; computer-assisted component)**.** *In the deterministic one-write model with public phase and four erasures,*

$$\boxed{\kappa_1(7,4) = 120.}$$

The lower bound uses the exhaustive five-case reduction and its separately checked finite DRUP certificates.

Proof. Any 119-label protocol implies a satisfying assignment in one of the five finite encodings. All five are refuted by the supplied DRUP traces. Conversely take the eight antipodal source pairs with seven-bit representatives

$$0,\ 7,\ 25,\ 30,\ 42,\ 45,\ 51,\ 52.$$

They give eight initial paired labels and 112 singleton labels. Their 112 distinct forced terminal pairs cover the full cube. The decoding and transition verifier checks every initial word, every one-write event, and every erasure mask of size at most four. This gives a valid protocol on 120 labels. □

By monotonicity of the admissible zero-error protocols in the erasure budget,

$$\kappa_1(7,5) \geq \kappa_1(7,4) = 120.$$

The same 120-label protocol corrects five erased coordinates because its phase-zero non-singleton fibers are antipodal and its terminal pairs have distance six. Consequently $\kappa_1(7,5) = 120$ as well. This implication permits the seven-coordinate proof of the stronger four-erasure problem to serve as the main finite certificate for both near-maximal erasure levels. The independent folded-cube closure enumeration retained in the supplement separately supports the structural claim $\Xi(C) \geq 6$ for complement-invariant 18-point configurations.

Trust boundary. The DRUP checker proves UNSAT for the five concrete DIMACS files. The source-to-CNF correspondence, the orbit completeness argument, and the collision–adjacency lemma are mathematical and source-level claims that require review on their own terms. The result is a reproducible computer-assisted finite theorem, without a Lean or Coq formalization of the entire reduction.

11 Unified optimality and the erasure plateau

Theorem 11.1 (Two adjacent erasure levels have the same optimal alphabet)**.** *For every integer $n \geq 7$ and either $q = n - 3$ or $q = n - 2$,*

$$\boxed{\kappa_1(n, q) = 2^n - \left\lfloor \frac{2^{n-1}}{n} \right\rfloor + \mathbf{1}_{\{n=7\}}.} \tag{19}$$

For all $n \geq 8$ the optimal value is supported by an analytic converse and explicit Hamming-prefix cubical constructions. The exceptional length $n = 7$ uses the finite computer-assisted theorem in Section 10. The minimum bit budget remains n in both levels.

Proof. For $n \geq 8$, Theorem 9.1 gives the same universal lower bound at $q = n-3$ as Theorem 5.2 gives at $q = n - 2$. Lemma 7.2 and the variational construction supply a protocol with $N - \lfloor 2^{n-1}/n \rfloor$ labels and even recover $n - 2$ erasures. Hence it is valid at both levels. At $n = 7$, Theorem 10.2 gives $\kappa_1(7, 4) = 120$. The same explicit protocol corrects five erasures, and monotonicity gives $\kappa_1(7, 5) \geq 120$. Thus both values equal 120. In all cases $2^{n-1} < \kappa_1(n, q) \leq 2^n$, so rounding the logarithm gives n bits. □

For $q \geq n-1$ the initial collision restriction makes the encoder injective, so $\kappa_1(n, q) = 2^n$. Thus the integer-label plateau in Theorem 11.1 is immediately adjacent to full-alphabet saturation. It is invisible when measuring only the rounded bit budget.

n	$\lfloor 2^{n-1}/n \rfloor$	$\kappa_1(n, n-2)$	$\kappa_1(n, n-3)$	Bits	Method
7	9	120	120	7	Five DRUP cases
8	16	240	240	8	Prefix cubes
9	28	484	484	9	Prefix cubes
10	51	973	973	10	Prefix cubes
11	93	1955	1955	11	Prefix cubes
12	170	3926	3926	12	Prefix cubes
13	315	7877	7877	13	Special suffix union
16	2048	63488	63488	16	Dyadic extremizer

Table 4: The exact one-write alphabet plateau. For $n = 4, 5, 6$ the $n - 2$ layer is known but the plateau theorem does not assert the same values at $n - 3$.

12 Related work and contribution boundaries

Zero-error coding with side information. Witsenhausen's zero-error side-information framework relates minimum message alphabets to graph colorings [1]. In the static case of our model, two binary words can share an encoding label precisely when their Hamming distance exceeds the erasure budget; the resulting chromatic number of a power of the binary cube is the appropriate baseline. Exact finite colorings, including Kokkala and Östergård's solution for the square of the eight-cube [4], demonstrate that these static invariants need not admit simple uniform formulas. More recent graph-entropy formulations of zero-error computation with side information further illustrate the breadth of the static characteristic-graph approach [12].

Our additional constraint is a consistent transition mapping for every specified blind coordinate assignment. Thus an initial color class cannot be optimized without regard to the labels available after a write.

Causal coding and rewritable storage. Witsenhausen also investigated the structure of finite-horizon real-time source coders under causal information constraints [9]. Ahlswede and Zhang's write-efficient memory model studies data rewriting through a transition cost and its associated rate–cost tradeoff [10]. These antecedents motivate the distinction between constraints on the encoder's information, the number of encoded states, and the cost of changing a physical representation. Our model charges the cardinality of a persistent state alphabet, makes the previously stored source coordinate unavailable to the online transition, and supplies the current unerased coordinates to the decoder. It does not optimize the Hamming distance between successive physical codewords.

Locally updatable codes and oblivious updates. Chandran, Kanukurthi and Ostrovsky introduced locally updatable and locally decodable codes that balance limited codeword modifications and recovery locality [11]. Nakkiran, Shah and Rashmi derived communication bounds and constructions for oblivious updates of stale storage nodes [3]. Both concern changes to encoded storage, but the latter counts refresh communication and the former counts locality/rewrite operations; neither has the identical one-write, zero-error, phase-aware state-alphabet objective studied here.

Specific contributions and their scope. Under the explicit assumption of one blind coordinate assignment, a public phase index, and zero-error recovery for all erasure patterns, we derive a high-erasure matching normal form and bound the multiplicity of forced successor edges by initial hypercube adjacencies. Combining these tools with explicit cubical constructions gives the exact state-alphabet plateau at the two adjacent erasure budgets $q = n - 3$ and $q = n - 2$. The exceptional seven-coordinate lower bound uses an exhaustive finite reduction and separately checkable propositional refutations. The statement concerns exact *label counts*; the rounded optimal bit budget remains n in the stated regime. The analysis does not establish a complexity classification for arbitrary update horizons or a general coding theorem for all dynamic-storage models.

The two costs κ_1 and R_1^{on} must be distinguished throughout. In particular, the two near-maximal erasure levels require n binary memory places despite a strict reduction in the number of distinct labels. The integer cardinality result therefore measures a fine-grained state-space saving inside a single bit-complexity plateau. Whether the exact saving itself translates into a practical implementation advantage depends on representation and indexing overhead; explicit schemes can rank states without storing a lookup table of exponential size when the structure admits a succinct representation. We do not claim measured systems throughput or energy savings.

The new exact value computation for $T = 1$, $q \in \{n - 3, n - 2\}$ is polynomial time in the length of the output written explicitly in binary, and the associated threshold comparison can use bit-length short circuits. This does not classify the complexity of optimizing the alphabet for arbitrary user-supplied (n, T, q), nor prove that a candidate-restricted geometric optimization has the same complexity as the unrestricted one.

13 Reproduction and limitations

The source package includes the fully specified analytic construction and enumerators from the V7 and V10 line of results, the V9 matching reduction, and the five concrete CNF/DRUP pairs from V12. The canonical command for the finite certified case is `bash verify_all.sh` inside the

V12 certificate folder. That command verifies SHA-256 hashes, tests exact-cardinality constraints, regenerates five CNFs byte-for-byte, checks all five proof traces, checks the seven-cube geometry and a 120-label witness, and tests rejection of damaged proofs. The whole-paper analytic proofs are written out in the main text; finite testing alone is not treated as a proof of their general quantifiers.

This manuscript is a research draft with independently reproducible finite certificates; it has not undergone external peer review. External refereeing should address a fresh review of the Hamming-prefix packing lemma (including the exceptional $n = 13$ construction), the collision–adjacency inequality for the $n - 3$ layer, and the combinatorial-to-CNF implications in the seven-dimensional finite argument. A second proof checker would further narrow the trust boundary. Originality should also be tested against the newest literature in zero-error dynamic coding and constrained graph colorings; the references cited here verify related resource formulations but do not by themselves establish priority.

14 Conclusions

A single blind overwrite creates constraints on the compatibility of initial state labels and their successors. At maximal nontrivial erasure budget the exact state alphabet admits a finite variational formula over antipodal source subsets. A geometric converse and Hamming-prefix constructions determine its optimum at all blocklengths. The next erasure level has the same closed-form optimum for all $n \geq 7$; the seven-coordinate exception is supported by a constructive protocol and independently checkable propositional proof traces. The matching normal form supplies a wider exact optimization framework in the high-erasure regime. The distinction between an integer state alphabet and the rounded bit budget is essential to the resulting plateau phenomenon.

A A finite-reduction proof ledger

A refutation of a propositional formula establishes the absence of assignments satisfying that formula. For the seven-dimensional theorem, the implication from a hypothetical short online protocol to a satisfying Boolean assignment is independent of the SAT search and is made explicit here.

Proposition A.1 (Complete coverage by the five case families)**.** *If a one-write, four-erasure protocol on $\mathbb{F}_2^7$ uses at most 119 state labels, there exists a set of nine pairwise disjoint initial word pairs, each at distance six or seven, whose selection variables satisfy one of* `B.antipodal`*,* `B.mixed`*,* `A.normal`*,* `A.special`*, or* `A.antipodal`*.*

Proof. Lemma 10.1 yields $a = 9$ and $e - D + \lceil \Xi/2 \rceil \leq 1$. Because both terms are nonnegative integers, either $\Xi = 0$ or $e = D$ and $\Xi \leq 2$. In the former case, translate and permute coordinates so that an antipodal selected pair is $\{0, 127\}$ if all pairs are antipodal, or normalize any selected distance-six pair to $\{0, 63\}$. These are the two B families.

In the latter case the radius-one neighborhood identity reads $|B_1(C)| = 144 - 2e - \Xi \leq 128$, hence $e \geq 8 - \Xi/2 \geq 7$. Since internal unit edges in a selected initial pair are impossible, e is the sum of cross-pair unit-edge counts. The inequality in Lemma 8.1 gives a nonnegative loss $e(P, Q) - |\mathcal{F}(P) \cap \mathcal{F}(Q)|$ for each two different selected pairs. The absence of triple disjoint predecessors implies $D = \sum_{P<Q} |\mathcal{F}(P) \cap \mathcal{F}(Q)|$. Therefore $e = D$ forces each pairwise loss to vanish. Because $e \geq 7$, some two distinct selected pairs have a cross-unit edge. A single cross-unit edge cannot create a common successor by Lemma 8.1, so their cross-edge count equals two, and they share two forced successor pairs. In Q_7, these strong collisions have exactly three isometry orbits; an enumerator checks all 512 legal initial pairs, all strong collisions, and all cube

translations and coordinate permutations, with orbit sizes 1,344, 224, 224. Their representatives are the three A seeds used in the finite encodings. This covers every hypothetical protocol. □

Boolean encoding map. For an admissible source pair p, the primary variable z_p is true exactly when p is selected. Exactly-nine cardinality constraints and at-most-one-per-endpoint constraints make the selected pairs a matching of size nine. The auxiliary u_v is defined bidirectionally by $u_v \leftrightarrow \bigvee_{p \ni v} z_p$, hence records membership $v \in C$. In a B case, the clauses $u_v \vee \neg u_a \vee \neg u_b$ for distinct cube neighbors $a, b \in N(v)$ enforce $\Xi(C) = 0$.

In an A case, for exterior v define the threshold indicators

$$t_{v,j} = \neg u_v \wedge [d_C(v) \geq j], \qquad j = 2, 3, 4.$$

The gate network enforces each threshold in both directions and allows at most two true indicators in total. This is exactly the condition $\Xi(C) \leq 2$: for an exterior vertex with $d_C(v) \leq 3$ its contribution is $d_C(v) - 1$, while if $d_C(v) \geq 4$ the first three indicators already exceed the global budget. The constraints $\neg z_p \vee \neg z_q$ for pair-pair combinations with $e(p, q) > |\mathcal{F}(p) \cap \mathcal{F}(q)|$ encode $e - D = 0$, because the pairwise losses are nonnegative. Unit clauses set the selected symmetry seeds. Thus every putative protocol yields an assignment satisfying its normalized instance; a sound UNSAT certificate for all five instances excludes such protocols.

Trusted boundary. The finite enumeration of orbit representatives and the direct encoding map are part of the mathematical reduction. The DRUP proof checker certifies five concrete CNFs, not the reduction itself. In the archive, `verification/verify_folded_cube_exception.py` independently checks the $q = n - 2$ folded-cube obstruction, while `verification/independent_v10_audit.py` checks the local forced-edge geometry for $n = 7, 8$. The certified CNF/DRUP pairs and their SHA-256 values are bundled under `v12_certificates/`; the complete reproduction command is `bash verify_all.sh` in that directory.

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